MLAPS: A Machine Learning based Second Line of Defense for Attack Prevention in IoT Network

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2022 IEEE 19th India Council International Conference (INDICON) | 978-1-6654-7350-7/22/$31.00 ©2022 IEEE | DOI: 10.1109/INDICON56171.2022.10039777

***Abstract*—Intrusion Prevention Systems (IPSs) are processes that are handled as add-ons to the Intrusion Detection Systems (IDSs) in order to effectively resist and prevent invasions in IoT devices. In this paper, a comparative study on various Machine learning (ML) based Intrusion Prevention System (IPS) has been proposed on UNSW-NB15 dataset by applying various ML methods for classification. For Intrusion Prevention System, two level classifiers for dropping of packets are implemented. Level 1 classifier comprises of Decision Tree (DT) whereas Level 2 classifier uses Random Forest (RF). The proposed IPS automatically adapts to system changes and further reduces training time and increases the accuracy of the system and acts as a lightweight Second Line of Defense System for an IoT Network. In this system, threshold value is calculated by finding the average time of execution of all of the data in the training set.**

***Index Terms*—IPS, Intrusion Prevention System, Machine Learning, Intrusion Detection System, IoT, Internet of Thing**

I. Introduction

Securing data, networks and communication among Internet of things (IoT) devices and with the Internet is quite onerous to protect IoT ecosystems. IoT has many challenges including constrained resources particularly in terms of memory and energy, low computing power, insecure operating systems, insufficient access controls, lack of transport encryption, unprotected network access and limited interfaces, pose a threat to IoT security [1]. The well-known Mirai Botnet [2] attack in 2016 led down many internet services and IoT devices worldwide such as IP cameras, printers, baby monitors, gateways and home routers. The defense strategy is mainly two lines of defence systems [3]:

A. First line of defence

Firewalls, as well as elements for access control, authentication, encryption and secure key management by using technologies like cryptography, Block-chain and Quantum Computing [4,5] are included in this layer of security.

B. Second line of defence

Intrusion detection systems (IDSs), Intrusion Prevention Systems (IPSs) and Intrusion Response systems (IRSs) are referred to as defensive security systems for invasions. Any software or hardware that identifies intrusions is referred as an Intrusion Detection System (IDS). Intrusion Prevention Systems (IPSs), sometimes known as inline IDS are an extension of IDS that aim to detect intrusions and then protect the system in real time. Intrusion Prevention Systems (IPS) and Intrusion Detection Systems (IDS) are worthless without Intrusion Response Systems (IRSs), which utilise protection mechanisms to evaluate system performance and even identify and address possible intrusions. The overview of the IDS and IPS is explained below.

1) Intrusion Detection System: Monitoring, detection and response are the three general components of a conventional IDS. The traffic flow behaviour is analysed by the monitoring component. Any questionable behaviour is noticed by the detection component which then tells the response component. The response component sends a message to the network administrator or raises an alert [6].

2) Intrusion Prevention System: IDS and IPS both share and even update a shared knowledge base while having slightly diverse roles.The IDS detects the intrusions and may generate a passive alert which further may be given as an input to IPS for taking further actions. IPSs are the extensions of IDSs [7] since they monitor systems or network activities and to stop/block intrusions by taking set of actions from prevention engine. IPSs, unlike IDSs, are installed in-line and can block intrusions before they are noticed by an user. For example, IPS can discard malicious packets, transmit alerts, reset the connection, fix transmission faults, clear out undesirable network and transport layer choices and so on [8]. The two stages of defensive security prevention and detection are as depicted in Figure 1. To maximise network security, they frequently cooperate or might be included into a single defensive system.

In this paper, an Intrusion Prevention System (IPS) has been proposed which uses two level classifiers for dropping of data packets in two phases. Phase 1 consists of models formation for level 1 classifier using Decision Tree (DT) and level 2 classifier using Random Forest (RF) and calculation of threshold value (θ). Phase 2 deals with splitting of testing dataset on the basis of threshold value (θ).

The paper is outlined as follows: The literature review on the major intrusion detection systems for IoT devices is described in Section 2. In Section 3, the proposed methodology

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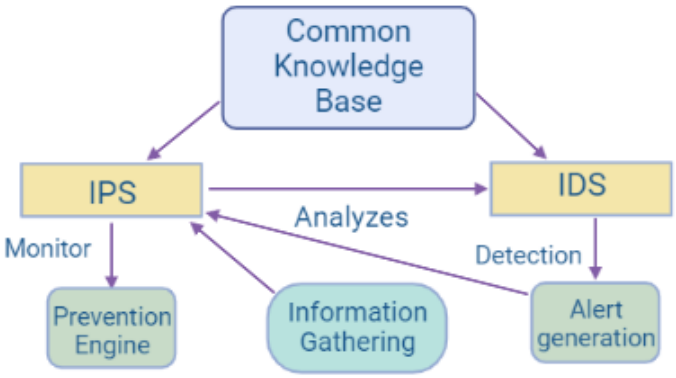


Fig. 1: IDS To IPS

along with the dataset are described. Section 4 illustrates the overview of the model used for analysis and section 5 deals with the obtained results or observations while section 6 concludes the paper.

II. Literature Review

In paper [9], IPS with reCAPTCHA Controller using information-based metrics for DDoS [10] attack on VANET prohibits bots from launching consistent automated attacks as well as many automated DDoS attacks. This method examines the randomness of user requests using information theory-based measures, such as entropy. Both frequency and entropy are utilised to determine the attack's vulnerability. DDoS attacks are carried out using AODV, Firecol and reCAPTCHA. It has been proven that reCAPTCHA performs well during DDoS attacks.

In paper [11], Intrusion Prevention Systems (IPSs) are focused on big data. The author presented a huge data-based hierarchical deep learning system to improve efficiency. It Captures both network traffic and content information using behavioural and content-functional capability. This model enhances the accuracy of fake attack categorization. SVM and DT were the algorithms used for implementation in this article. The independent test and the 52 cross-validation F test are used to assess IDS performance. The function extraction and selection method may consume too much storage space for large datasets. Storage for sparks For an efficient approach to role collecting, the MapReduce architecture is used.

In paper [12], according to Hybrid Machine Learning Network Intrusion Prevention System in Real-Time was capable of high performance and real-time categorization. This system employs two classifiers. Only if the Level 1 classifier can categorise packets as normal or attack with precision can it receive them at line speed. Level 2 classifier has no real-time processing limits, therefore it can take any leftover packets that were not classified as attack in level 1 and categorise them slowly but accurately. The dataset is the same in both.

In paper [13], Improving Cyber Security in advanced Internet of things (IoT) Using Intrusion Prevention Algorithm for IoT (IPAI) was discussed. The suggested technique focuses on the investigation and research of bandwidth attacks with a special focus on DDoS, which is a difficult to detect problem that degrades network performance. DDoS is a sort of assault in which a collection of attacker nodes attacks the target, preventing legitimate users from accessing the resources and network services. The proposed technique is based on the report generated by the IDS after analysing the forensic analysis report, which further leads to the prevention of DDoS assaults in the network.

In paper [14], an IDS/IPS (Intrusion Prevention/Detection System) named Dique that can identify and stop denial of service (DoS) assaults was developed using a deep learning algorithm. The information of recorded and identified packets can be graphically and textually displayed in real time via Dique's Graphical User Interface (GUI). Deep Feed Forward neural network with multiple layers is used in the proposed model for classifying DoS attacks. For training, the CICDDoS2019 Dataset was utilized. The Deep Feedforward technique has a 99.4% accuracy rate. To verify the functionality of Dique, the online application Diluvio was created and constructed.

In paper [15],author proposed a cutting-edge unsupervised intrusion prevention system (IPS) for vehicle Controller Area Networks that can identify and prevent assaults without altering the design of the electronic control units (ECUs) or requiring data that is only available to automakers. They have assessed which machine learning (ML) method can identify fuzzy and spoof attacks with the greatest degree of precision with the fewest amount of data bytes. It is the only approach that can reject attacking frames before damage is done and may be utilised on a low-cost Raspberry Pi. They have used three datasets from Hacking and Countermeasure Research Lab [16]. Detection results on these datasets are fuzzy dataset 99.29%, spoof dataset 99.24%, RPM guage spoof dataset %.

III. Methodology

In this proposed Intrusion Prevention System (IPS), two level classifiers for dropping of data packets are implemented. This two level classification is done in two phases. Phase one consists of models formation for level 1 classifier and level 2 classifier. Level 1 classifier comprises of Decision Tree (DT) whereas Level 2 classifier uses Random Forest (RF). Intrusion Prevention System (IPS) which automatically adapts to system changes and further reduces training time and increases the accuracy of the system. In this system, calculation of (θ) is dealt by first finding the execution time of each data point in the dataset then finding the average time. Phase two deals with splitting of testing dataset on the basis of threshold value (θ).

A. Dataset

The UNSW-NB 15 data set [17-21] was generated in the Australian Centre for Cyber Security's (ACCS) Cyber Range Lab which is a mix of genuine modern regular activities and synthetic contemporary threat behaviours. Analysis, Backdoors,Fuzzers, DoS, Exploits, Reconnaissance,Generic, Shellcode, and Worms are among the nine types of attacks. This dataset comprises of 49 features and total records in the

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TABLE I: Different intrusion prevention and detection approaches

S.No.	Ref.	Feature Selection	Classification techniques	Attack Model	Dataset
1	[9] Poongodi M	Not used	Information theory based metric	reCAPTCHA technique	Realtime
2	[11] Abdulrahman Saad Alqahtani	Not used	Hierarchical Deep Learning System	HIDS	CICIDS2017
3	[12] Wooseok Seo , Wooguil Pak	Uses each byte value without one-hot encoding	RF and DT	NIDS	UNSW-NB15 and CICIDS2017
4	[13] Abdulaziz Aldaej	Not used	Forensic analysis of IDS report	Not used	DDos Attacks
5	[14] Canola Garcia, J. F., Blandon, G. E. T.	CICFlowMeter tool	Multilayered Deep Feed Forward neural network	Not used	CICDDoS2019
6	[15] Paulo Freitas de Araujo-Filho	Not Used	One-Class Support Vector Machine (SVM) and Isolation Forest	Not used	Hacking and Countermeasure Research Lab datasets [16]

Dataset is 1,75,341 containing various attack and benign data.

TABLE II: Dataset Features

Feature type	**Feature Name**
Nominal	'proto', 'service', 'state', 'attack_cat'
Integer	'sbytes', 'sttl', 'sloss', 'swin', 'stcpb','dtcpb', 'dwin', 'trans_depth', 'ct_srv_src', 'dloss', 'ct_state_ttl', 'ct_dst_ltm', 'ct_src_dport_ltm', 'dttl', 'ct_dst_src_ltm', 'ct_dst_sport_ltm','ct_ftp_cmd', 'ct_flw_http_mthd', 'dbytes','ct_srv_dst'
Float	'dur', 'tcprtt', 'synack', 'ackdat'
Binary	'is_sm_ips_ports', 'is_ftp_login'

For Binary Classification, feature label has two labels as 19,488 records as normal data and 61685 records as attack data. For Multiclass Classification, feature attack_cat

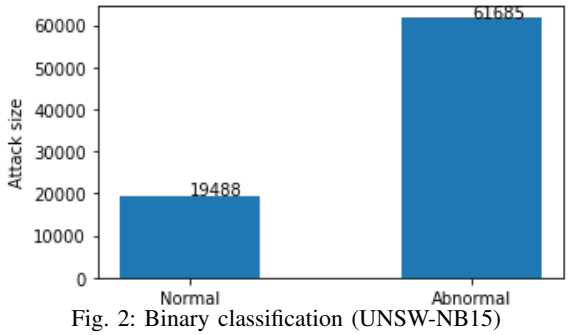


Fig. 2: Binary classification (UNSW-NB15)

has 9 labels Analysis(564), Backdoor(99), Dos(1,791), Exploits(16,187), Fuzzers Multi labels(1,731), Generic(39,496), Normal(19,488), Reconnaissance(1,703), Worms(114).

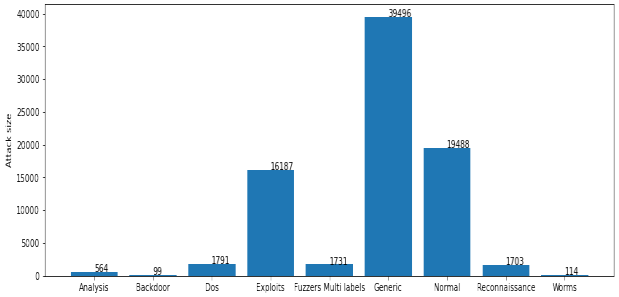


Fig. 3: Multiclass classification (UNSW-NB15)

B. Data Preprossessing and Normalization

Data must be cleaned up and prepared for a machine learning model in order for the model to be more accurate and efficient. It's essential to perform data preparation to clean the data and to make it ready for a machine learning model, which accuracy and efficiency of the model. As dataset might contain null or void values. Data Prepossessing moves from data collection, data formatting to data transformation. Normalization is responsible for scaling of data, as it makes data more of use for classification. Feature Selection is one of the major aspect to improve model accuracy and as well as makes the computational cost low. In this paper, for Feature Selection, Correlation matrix is used. Figures [4−6] are made to depict the correlation between the features.

Dataset's feature 'select' consists of values '-' so entire rows are deleted from dataset. Variant numeric data types are converted into single numeric data type.

Nominal/categorical data is dealt using **one-hot encoding** i.e., features which lie in this category are **proto**, **service**, **state**. Categorical parameters will prepare individual columns for every distinct labels in the above mentioned category.

Total features after encoding are 61. All numeric datatype features are normalized using **MinMaxScalar()** with range(0,1). The largest feature is scaled to one and the smallest feature to zero in another method of scaling data. The MinMax Scaler reduces the inputted data range, which is typically 0 to 1. Data is changed by scaling properties to a preset range. Without altering the underlying distribution's form, it modifies the values to fall inside a specified value range.

Binary labels are formed using **LabelEncoder()**, where as multiclass labels are formed using **one-hot-encoding** and **LabelEncoding**.

C. Feature Selection

Correlation matrix is formed and features with correlation value less than 0.3 are removed from dataset. This strategy is applied for binary data and multiclass data selection.

IV. Models used for analysis

A. Logistic Regression Algorithm

Logistic regression [22] evaluates the probability that an event, such as selected or not selected as set of independent variables are given. As the outcome is a probability, value of the dependent variable lies between 0 to 1 inclusive.

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B. Naïve Bayes Algorithm

The Naive Bayes (NB) [22] Classifier is used to solve complex classification problems. Naive Bayes (NB) refers to a class of supervised machine learning classification methods completely dependent on Bayes theorem. Although it is a simple categorization method, mostly used when the inputs are of large dimensional.

C. K Nearest Neighbor Algorithm

The supervised learning domain includes the K Nearest Neighbor (KNN) [22] method, which is used for both regression and classification but classification takes an upper hand. It is a versatile method that can be used to resample datasets and calculate absent values. It is most commonly used to find the class(integral) value or incessant value of a data point.

D. Decision Tree Algorithm

Decision tree (DT)[22] algorithm employs a divide and conquer technique by using a greedy search strategy to find the optimal break points within a tree. Now recursively, in a top-down way splitting of tree is proceeded further, till all or more than enough class labels are specified.

E. Random Forest Algorithm

Random Forests (RF) [22] are used for both regression as well as classification problems. It builds decision trees and for classification mean of the trees are used whereas the maximum vote strategy for regression. Both categorical as well as continuous types of datasets are well handled by Random Forest (RF) Algorithm.

F. AdaBoost Algorithm

The Boosting approach known as AdaBoost [22] algorithm, also identified as Adaptive Boosting, and is used as an Ensemble Method. Boosting works on the basis that students advance in stages. The weights are redistributed to each instance, with higher weights assigned to instances that were incorrectly classified.

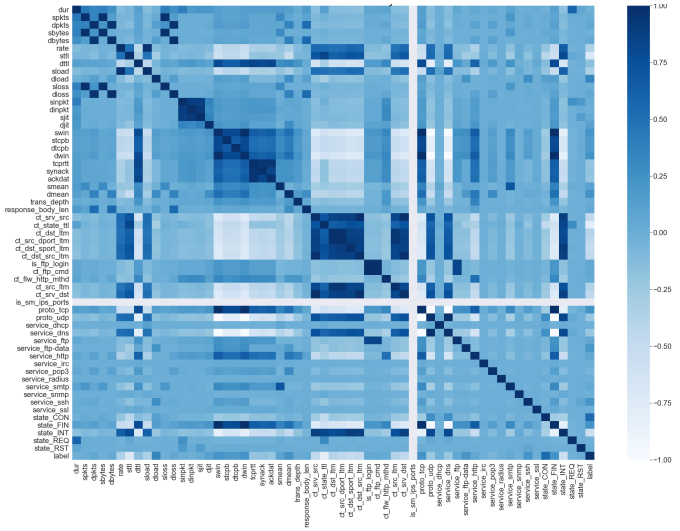


Fig. 4: Correlation matrix Binary Data

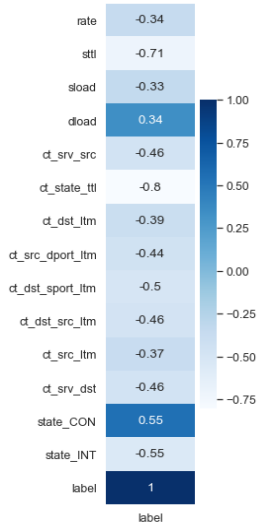


Fig. 5: Feature Selection Binary Data

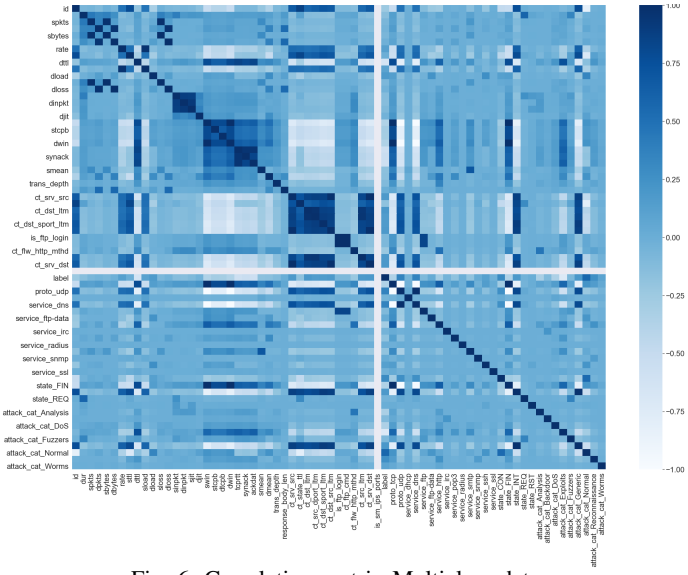


Fig. 6: Correlation matrix Multiclass data

G. Support Vector Machine Algorithm

Support Vector Machine (SVM) [22] is a supervised ML approach. It could be used to solve regression as well as classification problems. For the transformation of data method used is identified as kernel trick, Such transformation is necessary to identify the best output boundary in SVM.

V. Observation

Seven ML models are implemented for both binary classification as well as multi-class classification. For each model it's accuracy, precision, recall, f1-score and time to execute the model is noted. Since the concern is over the trade off between the accuracy and time to execute the model so models with high accuracy and less execution time is noted for both binary classification as well as multi-class classification. It is observed that Decision Tree takes less time compared to other

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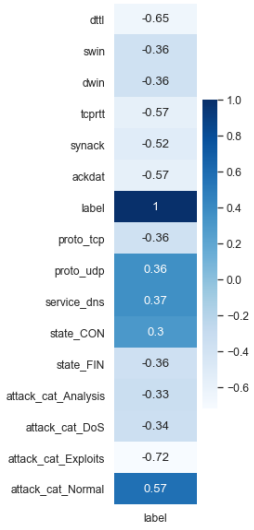


Fig. 7: Feature Selection Multiclass Data

models and also has high accuracy. Whereas Random Forest model has the highest accuracy.

A. Selection of Models

Binary classification and Multi-class classification on UNSW-NB15 is depicted by Table III IV.

TABLE III: Binary classification

Model	Accuracy	Precision	Recall	F1	Time(s)
LR	0.977	0.98	0.98	0.98	0.00267
NB	0.74	0.87	0.75	0.76	0.01101
KNN	0.982	0.98	0.98	0.98	4.27056
DT	0.980	0.98	0.98	0.98	0.00444
RF	0.984	0.98	0.98	0.98	0.04627
AdaB	0.981	0.98	0.98	0.98	0.07368
SVM	0.966	0.96	0.96	0.96	0.75300

TABLE IV: Multiclass classification

Model	Accuracy	Precision	Recall	F1	Time(s)
LR	0.974	0.97	0.97	0.97	0.00763
NB	0.951	0.95	0.95	0.95	0.04636
KNN	0.972	0.97	0.97	0.97	17.2074
DT	0.970	0.97	0.97	0.97	0.00661
RF	0.973	0.97	0.97	0.97	0.07719
AdaB	0.751	0.63	0.75	0.67	0.22904
SVM	0.973	0.97	0.97	0.97	1.26248

So for Intrusion Prevention System Models to be used are Decision Tree & Random Forest.

B. Intrusion Prevention System

In contrast, IPSs are said extensions of IDSs [7] since they monitor system or network activities to stop/block intrusions. IPS, unlike IDS, are installed in-line and can block intrusions before they are noticed. For example, IPS can discard malicious packets, transmit alerts, reset the connection, fix transmission faults, clear out undesirable network and transport layer choices and so on [8].

Proposed Intrusion Prevention System consists of two phases. Phase one consists of models formation for level one classifier Decision Tree and level two classifier Random Forest. Calculation of (θ) is dealt by first finding the execution time of each data point in the dataset then finding the average time. Phase two deals with splitting of testing dataset on the basis of threshold value (θ).

1) Phase 1 : Formation of models and calculation of threshold value: firstly, both models are trained using Decision Tree Random Forest. Threshold value (θ) is calculated by find the average time taken for classifying the training dataset.

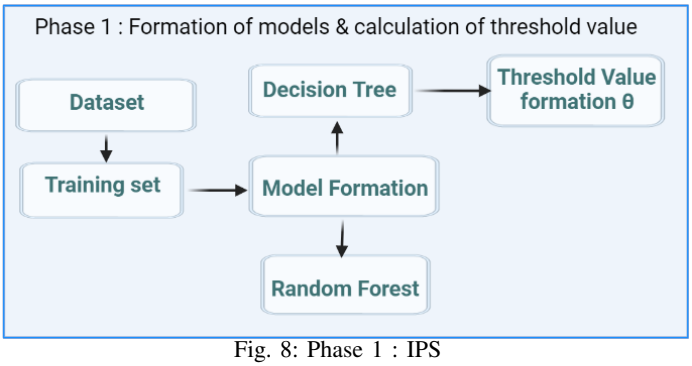


Fig. 8: Phase 1 : IPS

2) Phase 2 : Prevention - Dropping of packets: Now, Decision Tree Model is used for Level 1 classifier whereas Random Forest is used for Level 2 classifier. If the value is less than or equal to threshold value θ then L1 classifier is used otherwise stream is forward to L2 classifier. Both L1classifier , L2 classifier are used for prevention of attack.

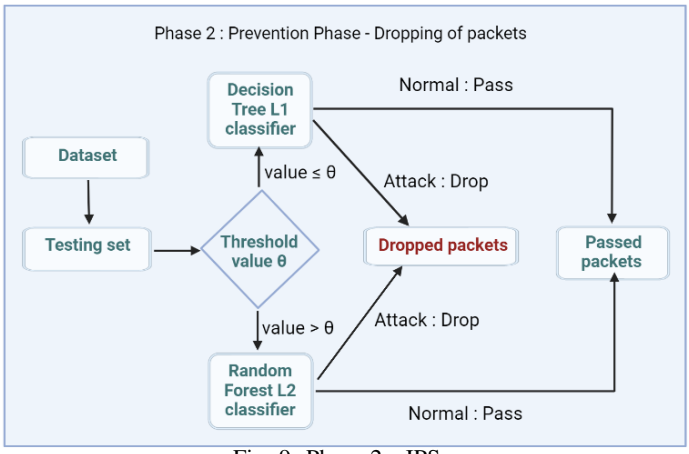


Fig. 9: Phase 2 : IPS

It is noted that accuracy for both the binary and multiclass classifiers results are slightly higher. Performance metrics for level classifiers(Decision Tree model and Random Forest model) for binary and multiclass classification is depicted below.

TABLE V: Binary classifiers

ML Model	Accuracy	Precision	Recall	F1
DT(L1)	0.981	0.98	0.98	0.98
RF(L2)	0.982	0.98	0.98	0.98

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TABLE VI: Multi-class classifiers

ML Model	Accuracy	Precision	Recall	F1
DT(L1)	0.970	0.97	0.97	0.97
RF(L2)	0.972	0.97	0.97	0.97

VI. Conclusion

In general, Prevention techniques may involve packet drop which leads to data loss. Models with higher accuracy require more computational memory which makes real-time prevention and detection less significant in terms of cost. Whereas a signature based methods have low computational cost but lower accuracy which are not ideal for the real-world applications [12].

The proposed hybrid model can be used as an IPS with improved training time, accuracy, and lower data loss. This research was not performed on an industrial scale with a broad attack span and is proposed as a future scope of this research work. Deep Learning techniques [10],[23] may also be explored as these techniques have also proven their efficacy in lower false positives with higher accuracy. In order to reduce data labeling, semi-supervised approach [24] may also be explored as a future scope of this work.

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